

Chapter 16 — Body Rhythms

Student Notes Workbook · 学习笔记工作册

Circadian · Ultradian · Infradian · Circannual rhythms

Name 姓名: _____

Class 班级: _____

Start date: _____

How to use this workbook 使用方法

1. **Open the courseware** (Ch16 Body Rhythms interactive page) and read one section at a time. 打开课件，一节一节地读。
2. **Fill in the blanks as you read** — the blanks are exactly the words you need for the exam. 边读边填空，挖掉的正是考试要写的关键词。
3. **Write your own sentences** in the open boxes — explaining in your own words is the best revision. 开放框里用自己的话写，这是最好的复习方式。
4. **Check with the answer key** on the last page, then tick the review tracker below. 用最后一页的答案自查，然后在下方打卡。每一页右上角标了答案所在的课件章节（ANSWERS → 课件 §X），找不到就翻到那一节。

Syllabus map — what the exam can ask 大纲考点

Syllabus	Content 内容	Workbook pages
3.1.4	The sleep-waking cycle: internal pacemakers (SCN) and exogenous zeitgebers 昼夜睡眠周期的内外调节	pp. 2–6
3.1.5	Infradian rhythms: the menstrual cycle; SAD, light therapy and drugs 月经周期与季节性情感障碍	pp. 7–10
3.1.6	Research into body rhythms 相关研究证据	throughout + p. 13
Exam format 考试形式: Paper 2 Section C — 12-mark essay: AO1 knowledge (4) + AO2 application (4) + AO3 evaluation (4) . Two real questions with planners: pp. 11–12.		

Review tracker 复习打卡

- ☐ First pass — all blanks filled 第一遍填完 Date: _____
- ☐ Answer key checked — corrections in pen 对过答案并用笔订正 Date: _____
- ☐ Both essay planners written out 两道真题框架写完 Date: _____
- ☐ Brain dump page done from memory 合上笔记完成自测页 Date: _____
- ☐ Final review before exam 考前最后过一遍 Date: _____

Definition 定义

Body rhythms are natural, _____ (recurring 重复发生的) cycles of biological processes in the body.

The four types — sorted by duration 按周期长短分四类

Type	Duration 周期长短	Example 例子
Circadian (昼夜节律)	_____ (约 24 小时)	The sleep-wake cycle
Ultradian (超昼夜节律)	Shorter than 24 hours	_____ (睡眠中的睡眠周期/阶段)
Infradian (亚昼夜节律)	_____ (超过 24 小时)	The menstrual cycle (≈ 28 days); SAD is also infradian
Circannual (年度节律)	About one year	_____ (动物一年一度的行为)

Memory aid 记忆法

ULTRA = shorter ("ultra-fast" cycles happen many times a day) · **INFRA** = longer ("infra-slow", slower than a day). **Circadian** comes from Latin *circa* (about) + *diem* (day) = about a day.

Zeitgebers 授时因子

An **exogenous zeitgeber** (外部授时因子) is an _____ (external 外部的) event that helps to _____ (reset 重置) our biological clock. The **main** zeitgeber is _____ (光). Three others named in the textbook: _____, _____ and _____ (温度、运动、吃饭时间).

In your own words 用自己的话写

A friend says: "Body rhythms don't matter, I can sleep whenever I want." Using one example from the table, explain in 2–3 sentences why rhythms matter.

Key terms 关键词填空

The **suprachiasmatic nucleus (SCN)** (视交叉上核) is a cluster of neurons located in the _____ (下丘脑). Its role is to act as a _____ (时钟/计时器) that maintains the sleep-wake cycle — the body's **internal pacemaker** (内部起搏器).

The **pineal gland** (松果体) responds to darkness by secreting _____ (褪黑素), a hormone which regulates the circadian rhythm.

The light → sleep pathway 通路填空

DAY 白天	Light in the blue-green spectrum is received by retinal cells (视网膜细胞) → detected by the _____ → the SCN tells the pineal gland to _____ (抑制 suppress) melatonin → we stay _____ (清醒).
NIGHT 黑夜	Darkness falls → less light detected → the SCN signals the pineal gland to _____ (分泌 produce) melatonin → melatonin is released into the bloodstream and promotes _____ (困倦) → we sleep.

§4 · How the cycle varies 节律的三类变化**(1) Across the lifespan 一生的变化 — melatonin levels**

Newborns <3 months	Little melatonin of their own — received through _____ (母乳)
3 months → puberty	Levels continue to _____ (上升)
Puberty 青春期	Levels _____ (下降 decline) — explains why teenagers _____ (熬夜晚睡)
Adulthood to ~40	Levels _____ (平稳 even out)
After ~40 → old age	Levels steadily _____ — may explain why older people _____ (醒得更早)

(2) Between individuals 个体差异

"_____ people" rise early and are alert in the morning; "_____ people" (早鸟型 / 夜猫型) rise later and are alert late.

(3) Lifestyle 生活方式

Three disruptors named in the textbook: _____, _____ and _____ (轮班、旅行时差、咖啡因) — caffeine interferes with _____ (褪黑素) function.

In your own words 用自己的话写

Why does a teenager find it hard to fall asleep before midnight? Use the words *melatonin*, *puberty*, *decline*.

Study 1 — DeCoursey et al. (2000) 花栗鼠研究

Procedure: free-living eastern chipmunks in the Allegheny Mountains (US) wore _____ (无线电项圈) .
30 chipmunks had surgical _____ to the SCN; **24** had sham surgery; **20** were intact controls.

Findings: significantly more SCN-lesioned chipmunks were killed by _____ (黄鼠狼) in the first 80 days. They showed increased _____ (夜间在洞穴中的活动) , which attracted predators.

Study 2 — Ralph et al. (1990) 仓鼠 SCN 移植

Procedure: an SCN from a _____ (供体 donor) hamster was transplanted into a hamster whose own SCN had been _____ (损毁 destroyed) .

Findings: the transplant not only _____ (恢复 restored) the circadian rhythm — the recipient _____ (采用了供体的节律) .

Study	What it shows 说明了什么	One weakness 一个弱点
DeCoursey (2000)		
Ralph (1990)		

Two cautions when using this research 两点谨慎

- 1. Animal research:** chipmunks and hamsters are not humans — be cautious when _____ (泛化 generalising) to the human sleep-wake cycle.
- 2. Other pacemakers exist:** _____ (外周振荡器) are biological clocks in organs and tissues around the body (e.g. regulating _____ (肝脏与消化)), synchronised with the SCN.

Wider issue — animal ethics 动物伦理

DeCoursey exposed chipmunks to _____ (被捕食) after lesioning their SCN. Animal research demands a _____ (成本收益分析) before animals are put at risk. Write one sentence: is this research justified?

Removing the zeitgebers: what happens? 拿走授时因子会怎样

Study	Key findings 关键结果 (填空)
Siffre (1962) case study · French Alps cave	Two months alone in a cave _____ (130 米) underground — no light, no watch. When told to resurface on 14 September he thought it was _____ (8 月 20 日). His free-running cycle was about _____ (24 小时 30 分).
Siffre (1972) Midnight Cave, Texas	Six months underground. Twice he adjusted to a _____-hour sleep-wake cycle: _____ (36) hours awake + _____ (12) hours asleep.
Aschoff & Wever (1962) WWII bunker · 3–4 weeks	_____ (85%) of participants adjusted to a sleep-wake cycle of _____ (25–27 小时) — agrees with Siffre.
Folkard et al. (1985) 12 students · speeding clock	Isolated with only a clock, secretly sped up. Students _____ (能同步) to 23-hour days but _____ (无法适应) 22-hour days — their cycle averaged _____ (25.48 小时). Conclusion: the internal clock _____ (抵抗极端外部改变).
Skene & Arendt (2007) blind participants	Partially sighted people have a _____ (正常) cycle; fully blind people with no light perception suffer daytime _____ (嗜睡), napping and poor night sleep.

Phase shifts 相位移动 — Boivin et al. (1996)

↓ Phase delay 相位延迟	↑ Phase advance 相位提前
Shifting the pacemaker to a _____ (更晚) time — e.g. travelling _____ (向西)	Shifting the pacemaker to an _____ (更早) time — e.g. travelling _____ (向东)

Put the logic together 串逻辑 (考试就按这个写)

- ① Remove light (Siffre; Aschoff & Wever) → the clock free-runs at about _____ hours → so light normally _____ (校准 resets) it to 24h.
- ② But the clock is not a slave to external cues (Folkard) → it has strong _____ (自主性).
- ③ No light perception → desynchronised cycle (Skene & Arendt) → light is _____ (必需品 essential).

Conclusion sentence to memorise 背下来: Internal pacemakers and exogenous zeitgebers _____ (共同作用 work together) to maintain a normal sleep-wake cycle.

(Application (应用价值) 是 AO3 最稳的采分方向。填空 = 考试能直接用的答案。)

Problem	What the research suggests 研究给出的解决方案 (填空)
Shift work 轮班工作	Shift workers risk workplace _____ (事故) through lack of alertness. Using _____ (工作场所人工照明) can reduce circadian disruption.
Jet lag 时差反应	_____ your sleep schedule _____ (在出发前逐步调整) to adjust to the new time zone.
Modern devices 电子设备	Screens produce artificial light that interferes with _____ (褪黑素抑制) . Recommendation: fit _____ (滤光/夜间模式) to reduce blue light.
Emotional well-being 情绪健康	Reduced daylight in winter can cause low energy and depression — this is _____ (季节性情感障碍 SAD) (see p. 9).

§8 part 1 · The menstrual cycle — the six steps 月经周期六步

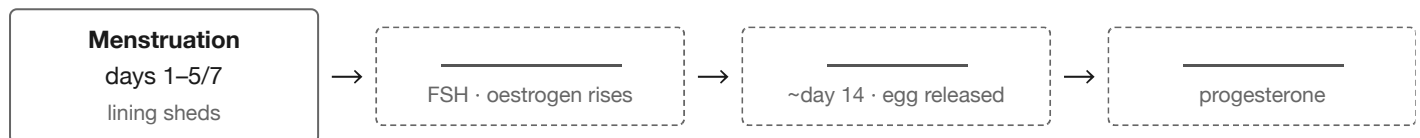
The cycle is about _____ (28) days and prepares the body for _____ (怀孕) . Fill the key hormone at each step:

1. **Menstruation** (月经期) : the lining of the _____ (子宫) sheds — lasts about _____ (5-7) days.
2. The **pituitary gland** releases _____ (卵泡刺激素) (FSH), which stimulates the ovaries to produce eggs.
3. _____ (雌激素) (oestrogen) is lowest on day 1 and rises as eggs mature.
4. **Ovulation** (排卵) at about day _____: a mature egg is released into the fallopian tube.
5. _____ (孕酮) (progesterone) is released to prepare the uterine lining for implantation.
6. If the egg is not fertilised, oestrogen and progesterone levels _____ (下降) → the lining is shed → the cycle repeats.

The command chain 指挥链填空

The internal pacemaker of the menstrual cycle is the **hypothalamic-pituitary-gonadal (HPG) axis**. It works as a _____ (反馈回路 feedback loop) at each stage:

1. The _____ (下丘脑) releases GnRH (gonadotropin-releasing hormone).
2. This stimulates the _____ (垂体) to secrete _____ (FSH) and _____ (黄体生成素) (LH).
3. FSH triggers the development of eggs and the production of _____ (雌激素) .
4. LH triggers the release of the egg during _____ (排卵) .
5. After ovulation, _____ (孕酮) prepares the uterine lining for implantation.

Diagram – label the cycle 图形标注

(Fill the four boxes using English terms (用英文术语填) : the first box is done for you.)

Hormones and mood – PMS and PMDD 激素与情绪

- **PMS** (经前综合征) : mood _____ (波动) and _____ (易怒) in the days before menstruation.
- **PMDD** (经前烦躁障碍) : an _____ (极端的 extreme) form of PMS with symptoms of _____ (抑郁) .
- Other effects: problems maintaining _____, _____ problems, changes in _____, increased pain sensitivity (注意力、记忆、食欲) .

In your own words 用自己的话写

Explain to a 12-year-old why the menstrual cycle is called a "feedback loop". Use the HPG axis words.

Evidence FOR internal pacemakers 支持内部起搏器的证据

Evidence	Key detail 关键细节
Hormonal science	The cycle is defined by the rise and fall of _____ and _____ (雌激素与孕酮); first tracked through the cycle by Georgeanna & Howard Jones.
The combined pill 复合避孕药	Contains _____ (合成激素) which suppress _____ (排卵) — highly effective.
Moos et al. (1969)	Pain and _____ (焦虑) changed with the _____ (周期阶段) — consistent across _____ (15) females.
Saglam & Basar (2019)	720 women; _____ (49%) had PMS, with higher _____ (愤怒) scores. Weakness: correlational — no _____ (因果).

Evidence FOR exogenous zeitgebers 支持外部授时因子的证据

Study	Key detail 关键细节
Lin et al. (1990)	16 females with _____ (48) -day cycles; a bedside light on days _____ (13–17) shortened cycles to _____ (33) days → light is a zeitgeber here too.
Labyak et al. (2010)	_____ (68) shift-working nurses; _____ (53%) reported menstrual changes.
Ahlborg et al. (1996)	Swedish _____ (助产士) working shifts had reduced _____ (生育力).
McClintock (1971)	_____ (135) dorm students: roommates and close friends showed similar dates of menstrual _____ (onset 开始) — "menstrual synchrony".
Stern & McClintock (1998)	Pheromone pads (信息素棉垫) from 9 donors rubbed onto the upper lip of 20 women: _____ (68%) aligned closer to the _____ (供体) 's cycle.
Ziomkiewicz (2006) 矛盾证据 contradictory	No synchrony in Krakow dorms (_____ (18) pairs, _____ (21) triplets); argues there is no _____ (进化优势) — it would only increase mate competition.

Wider issue — nature vs nurture 先天与后天

Infradian research sits on the _____ (先天 nature) side. But low mood may also come from _____ (社会条件 social conditions) (e.g. gender inequality and stress). Write the conclusion in your own words:

Definition & symptoms 定义与症状

SAD is a _____ (心境 mood) disorder: episodes of major depression during _____ (秋冬季节) .

Learn these five symptom keywords 背下五个症状关键词:

- ① Low _____ (情绪) and hopelessness ② _____ (疲倦与嗜睡) ③ Lack of _____ (动力与兴趣) ④ Changes in _____ (食欲) ⑤ (AO2 hook) symptoms return _____ (每年冬天) — a 12-month cycle.

Two biological mechanisms 两大机制 (2025 真题 AO1 采分点)

Mechanism	How it works 填空
 Melatonin 褪黑素	Winter darkness → the body produces melatonin for _____ (更长时间) → individuals feel _____ (疲倦与嗜睡) .
 Serotonin 血清素	Serotonin is known to _____ (减少 reduce) in low light → less mood regulation → _____ (情绪低落) . One theory: people with SAD show increased _____ (再摄取 reuptake) of serotonin.
Root cause 根本原因: reduced exposure to daily _____ → disruption to the _____ (昼夜睡眠觉醒周期) .	

Research evidence 研究证据 (填关键数字)

Study	Key finding 关键结果
Michalak & Lam (2002)	22 studies: latitude–SAD correlation $r =$ _____ (0.66) . Caution: _____ (迁移是个混淆变量) .
Magnusson & Axelsson (1993)	Icelanders who migrated to _____ (加拿大曼尼托巴) had _____ (更低) SAD rates.
Qin et al. (2015)	_____ (8) macaques in reduced daylight → huddling, reduced movement, weight loss — reversed by _____ (抗抑郁药) .
Golder & Macy (2011)	_____ (5.09 亿) tweets from _____ (84) countries: shorter days → _____ (更少积极情绪) .
Ma et al. (2006)	593 participants: _____ (86) kcal/day more in autumn vs spring.
Danilenko et al. (1994) (矛盾证据)	Higher daytime winter _____ (褪黑素) in SAD patients — but other studies failed to _____ (复制 replicate) → melatonin evidence is mixed.
Levitan et al. (1998)	Double-blind RCT: serotonin _____ (激动剂 agonist) drugs reduced sadness in _____ (14) female SAD patients.

Light therapy 光照疗法 — the numbers 数字填空

Brightness	About _____ (10,000 勒克斯)
Duration	_____ (20–60) minutes daily, autumn & winter
Timing	In the _____ (早晨), just after waking (Lewy et al. 2007 — corrects the winter _____ (相位延迟 phase delay))
How it works	Thought to _____ (减少) melatonin and _____ (增加) serotonin — although the mechanism is unknown.

Evidence for light therapy 光疗证据

Study	Key finding
Rosenthal et al. (1984)	Bright light morning + evening for two weeks → depression ratings _____ (下降); dim light → ratings _____ (上升) → effect depends on _____ (亮度 luminosity).
Golden et al. (2005)	_____ (元分析 meta-analysis) of RCTs: bright light significantly reduced depressive symptoms; dawn light ≈ _____ (抗抑郁药物试验效果).
Eastman et al. (1998)	_____ (96) patients, 3 groups; morning bright light > placebo (sham _____ (负离子发生器)) — but withholding treatment raises _____ (伦理) issues.

SSRIs 选择性血清素再摄取抑制剂 — mechanism first 先填机制

SSRIs _____ (阻断 block) the reuptake of serotonin by binding to the transporters → more serotonin stays in the _____ (突触间隙) → mood is _____ (提升 elevated).

机制与 SAD 的血清素理论严丝合缝 —— “病因研究指导治疗” 是现成的 AO3 点。

Study	Key finding
Lam et al. (1995)	68 patients, fluoxetine vs placebo: _____ (59%) % vs _____ (39%) % achieved ≥50% symptom reduction after _____ (5) weeks.
Moscovitch et al. (2004)	_____ (187) patients; _____ (舍曲林 sertraline) (50–200 mg/day) for 8 weeks beat placebo; drop-out very low.
Lam et al. (2006)	Light therapy vs fluoxetine head-to-head: _____ (同样有效 no difference) — but light therapy was _____ (更快).

Limitations checklist — tick the ones you can explain without looking 打勾自测

- ☐ Side effects (headaches, nausea, eyestrain; not for photosensitive medication, glaucoma/cataracts)
- ☐ Time commitment — sitting at a light box every morning
- ☐ No long-term research — recurrence unknown
- ☐ Individual variation in severity and response
- ☐ Mechanism unknown
- ☐ SSRIs: side effects, serotonin syndrome (rare), take weeks to work

The question. Noah has been diagnosed with seasonal affective disorder. Every winter he feels depressed. Noah's symptoms include lacking the motivation to see his friends, or go to the gym. He also sleeps more during the winter months. Noah does not eat as much food in the winter months as he does not feel hungry.

Evaluate infradian rhythms as an explanation of Noah's seasonal affective disorder. You must make reference to the context in your answer. (12 marks)

Plan your answer here — AO1 knowledge (4 marks)

- Define: an infradian rhythm lasts _____ hours; SAD is infradian because it follows a _____ - month cycle.
- Mechanism 1: winter darkness → more _____ → tiredness and oversleeping.
- Mechanism 2: less sunshine → less _____ → depression.
- Write one more AO1 point of your choice:

Apply to Noah — AO2 application (4 marks) 提示：每句都要写 "Noah"

Noah's symptom (from the question)	Which mechanism explains it? 你来配对
Feels depressed every winter	
Sleeps more in winter months	
Lacks motivation to see friends / gym	
Does not eat as much / not hungry	

Evaluate — AO3 (4 marks) 支持与反驳都要写

Support 支持：

- Wehr et al. (2001): people with SAD produce a biological signal of seasonal change, like other _____ (哺乳动物) that change behaviour seasonally.
- Application value: the explanation led to _____ (光疗) as a treatment.

Challenge 反驳：

- Sigurðardóttir & Grigariaviciuté (2018): being an _____ (夜猫型 evening-type) person is related to SAD — maybe it is the _____ (昼夜 circadian) rhythm, not infradian, that matters.
- Infradian explanations ignore _____ (社会因素) (e.g. seeing fewer friends in winter).

Mini-conclusion (write it yourself):

Timing discipline 时间控制

25 minutes total: AO1 ≈ 8 min · AO2 ≈ 5 min · AO3 ≈ 10 min · conclusion ≈ 2 min. Application alone is capped at 4 marks — **do not write a whole essay of description.** 不要只写知识点，别忘了评价。

The question. Liesel has got a new job in which she has to work during the night. She finds that she is very tired at work, even though she has slept during the daytime. Liesel thinks this is because she has not slept very well when she gets home from work in the morning. She finds it often takes her a few hours to get to sleep. Liesel often wakes up in the middle of the afternoon. She finds she cannot get back to sleep even though she does not need to get up for work.

Assess the role of internal pacemakers in the regulation of Liesel's sleep-wake cycle. You must make reference to the context in your answer. (12 marks)

Plan your answer here — AO1 knowledge (4 marks)

- The sleep-wake cycle is a _____ (昼夜 circadian) rhythm (24 hours).
- The SCN detects light/dark through information from the _____ (视神经 optic nerve) .
- Darkness → SCN stimulates the _____ (松果体) → melatonin production.
- Extra mark-winning detail: the SCN also regulates _____ (体温 body temperature) on a 24-hour cycle — lower at night, higher in the day.

Apply to Liesel — AO2 application (4 marks)

What happens to Liesel	Explain with the SCN / melatonin 你来解释
Very tired at work (night)	
Takes hours to fall asleep in the morning	
Wakes mid-afternoon and cannot get back to sleep	

Hint 提示: morning light + high daytime body temperature are the keys to the last two rows.

Evaluate — AO3 (4 marks)

Support 支持:

- Ralph et al. (1990): hamsters changed to a _____ (21) -hour cycle when implanted with an SCN from _____ (天生 21 小时节律的仓鼠) — the pacemaker drives the cycle.
- Application: Liesel can help herself — e.g. _____ (遮光窗帘 blackout curtains) in the daytime to allow melatonin production.

Challenge 反驳:

- Skeldon et al. (2023): preferred sleep timing is due to _____ (内部与外部 both) internal and external factors — pacemakers alone do not explain her cycle.
- Generalisability: much research uses _____ (个案研究 case studies) (e.g. Siffre) — may not apply to Liesel.

Mini-conclusion (write it yourself):

| Brain dump — close the courseware, use only the prompts 合上课件，只看提示词

- The four rhythm types + one example each:

- The light → sleep pathway (5 steps):

- Five zeitgeber studies (Siffre · Aschoff & Wever · Folkard · Skene & Arendt · DeCoursey/Ralph) — one finding each:

- Menstrual cycle: six steps + HPG axis chain:

- SAD: symptoms + two mechanisms:

- Treatments: light therapy numbers + SSRI mechanism:

| Answer key 答案速查（先做完再看！）

p.2 — recurring · circadian $\approx 24\text{h}$ · ultradian: sleep stages (cycles within sleep) · infradian: $>24\text{h}$ · circannual: hibernation & migration · zeitgeber def: external event that resets the clock · light; temperature, exercise, mealtimes.

p.3 — hypothalamus · clock/timer · melatonin · SCN · suppress · awake/alert · produce · drowsiness · breastmilk · rise · decline, stay up late · even out · decline, wake earlier · morning · evening · shift work, travel (jet lag), caffeine · melatonin.

p.4 — radio collars · lesions · weasels · nocturnal movement (in dens) · donor · destroyed · restored · adopted the donor's cycle (same rhythm as donor) · generalising · peripheral oscillators · liver function & digestion · predation · cost-benefit analysis.

p.5 — 130 m · 20 August · 24h 30min · 48 · 36 · 12 · 85% · 25–27 hours · synchronised · failed to adjust · 25.48 hours · resists extreme external shifts · normal · sleepiness · later · west · earlier · east · 25 · resets · autonomy · essential · work together.

p.6 — accidents · artificial light (workplace) · gradually shift ... before travelling · melatonin suppression · light filters · SAD (seasonal affective disorder) · 28 · pregnancy · uterus · 5–7 · FSH · oestrogen · 14 · progesterone · fall.

p.7 — feedback loop · hypothalamus · pituitary · FSH · LH · oestrogen · ovulation · progesterone · boxes: Follicle(lar) phase ·

Ovulation · Luteal phase · swings · irritability · extreme · depression · attention, memory, appetite.

p.8 — oestrogen · progesterone · synthetic (forms of) hormones · ovulation · anxiety · cycle phase · 15 · 49% · anger · causation · 48 · 13–17 · 33 · 68 · 53% · midwives · fertility · 135 · onset · 68% · donor · 18 · 21 · evolutionary benefit · nature · social conditions.

p.9 — mood · autumn & winter · mood · tiredness & oversleeping · motivation/interest · appetite · every winter · longer · tired/oversleep · reduce · low mood · reuptake · light · circadian sleep-wake cycle · 0.66 · migration (confounding) · Manitoba · lower · 8 · antidepressants · 509 million (509m) · 84 · less positive emotion · 86 · melatonin · replicate · agonist · 14.

p.10 — 10,000 lux · 20–60 min · morning · phase delay · reduce · increase · decreased · increased · luminosity · meta-analysis · antidepressant medication trials · 96 · negative-ion generators · ethics · block · synaptic gap · elevated · 59 · 39 · 5 · 187 · sertraline · no difference (equally effective) · quicker/faster.

p.11 — more than 24 · 12 · melatonin · serotonin · mammals · light therapy · evening-type · circadian · social factors.

p.12 — circadian · optic nerve · pineal gland · body temperature · 21 · hamsters with naturally 21-hour cycles · blackout curtains · both · case studies.

Final self-check 最后自问：如果我现在合上这本笔记，能在 25 分钟内写出 Noah 或 Liesel 的 12 分答案吗？
 如果不确定——回到课件 §13 再看一遍官方评分标准，然后重做 p.11 或 p.12。